



STEPCONE-2025

Standard Operating Procedure (SOP)

Name of the Event: CNC-Codethon

Conducted By: -Department of MECHANICAL

Expected No of Registrations: 30 teams

About the Event:

The CNC Codethon is a computer numerical control (CNC) programming competition that challenges participants to design and program CNC machines to manufacture complex parts. The event aims to promote innovation, creativity, and expertise in CNC programming, while providing a platform for participants to showcase their skills and learn from others. Roles & Responsibilities

Details of the event:

Round 1: Simple

- Challenge: Design and program a CNC machine to manufacture a simple part, such as a rectangular block with holes and pockets.
- CNC Requirements: Participants will need to write a CNC program to manufacture the part using a milling machine or lathe.

Venue	Duration	Faculty Coordinator	Student Coordinator
CAD Lab	3hrs	Dr. B.V.Suresh Dr. Ch.Vinodh Babu	D.V.S. Sushmitha B. Durga Prasad K. Satish P. Pawan Kumar S. Kowshik

Round 2: Medium

- Challenge: Design and program a CNC machine to manufacture a medium-complexity part, such as a bracket with curved surfaces and intricate features.
- CNC Requirements: Participants will need to write a CNC program to manufacture the part using a milling machine or lathe, incorporating techniques such as contouring and pocketing.

Round 3: Complex

- Challenge: Design and program a CNC machine to manufacture a complex part, such as a robotic component with intricate mechanisms and assemblies.
- CAD Requirements: Participants will need to create a 3D CAD model of the part using advanced features such as assemblies, mechanisms, and dynamic simulations.
- CNC Requirements: Participants will need to write a CNC program to manufacture the part using multiple CNC machines, incorporating techniques such as 4-axis machining and robotic assembly.

Guidelines of the Event:

1. Participants are expected to solve the given task within the given time.
2. Each round will be subject to a time limit.

Evaluation and Result:

The judges will evaluate the participants' performance in each round, and the final result will be determined based on the marks obtained by the participants in each round.



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